

(*Cuerda 1, amarilla*)

(*m=2kg*)

```
In[*]:= datafrecond1 = {{1 / 1.68, 12.4}, {1 / 0.84, 25.2}, {1 / 0.56, 36.5}, {1 / 0.42, 46.3},  
                      {1 / 0.336, 57.1}, {1 / 0.28, 73.2}, {1 / 0.24, 86.1}, {1 / 0.21, 101.9}};
```

(*a es 1/lambda*)

```
In[*]:= datafit1 = NonlinearModelFit[datafrecond1, {v * a + b}, {v, b}, a]  
          |ajusta a modelo no lineal
```

Out[*]=

FittedModel[$-1.51429 + 21.038 a$]

```
In[*]:= datafit1[{"BestFit", "ParameterTable"}]
```

Out[*]=

		Estimate	Standard Error	t-Statistic	P-Value
{ -1.51429 + 21.038 a,	v	21.038	0.641362	32.8021	5.34019×10^{-8}
	b	-1.51429	1.92781	-0.785496	0.462045

```
In[*]:= (*m=1.5kg*)
```

```
In[*]:= datafrecond5 = {{1 / 1.68, 11.3}, {1 / 0.84, 20.8}, {1 / 0.56, 28.7}};
```

```
In[*]:= datafit5 = NonlinearModelFit[datafrecond5, {v * a + b}, {v, b}, a]  
          |ajusta a modelo no lineal
```

Out[*]=

FittedModel[$2.86667 + 14.616 a$]

```
In[*]:= datafit5[{"BestFit", "ParameterTable"}]
```

Out[*]=

		Estimate	Standard Error	t-Statistic	P-Value
{ 2.86667 + 14.616 a,	v	14.616	0.775959	18.8361	0.0337662
	b	2.86667	0.997775	2.87306	0.213234

```
In[*]:= (*Cuerda 2, roja*)
```

(*m=1kg*)

```
In[*]:= datafrecond2 = {{1 / 1.68, 8.2}, {1 / 0.84, 17.2},  
                      {1 / 0.56, 25.5}, {1 / 0.42, 35.5}, {1 / 0.336, 42.5}, {1 / 0.28, 54.1}};
```

```
In[*]:= datafit2 = NonlinearModelFit[datafrecond2, {v * a + b}, {v, b}, a]  
          |ajusta a modelo no lineal
```

Out[*]=

FittedModel[$-1.04 + 15.1392 a$]

```
In[*]:= datafit2[{"BestFit", "ParameterTable"}]
```

Out[*]=

		Estimate	Standard Error	t-Statistic	P-Value
{ -1.04 + 15.1392 a,	v	15.1392	0.403486	37.521	3.01299×10^{-6}
	b	-1.04	0.935328	-1.11191	0.328499

```
In[*]:= (*m=1.5kg*)
```

```
In[*]:= datafrecond3 = {{1 / 1.68, 10.5}, {1 / 0.84, 21},  
                      {1 / 0.56, 30.8}, {1 / 0.42, 42.1}, {1 / 0.336, 51.7}, {1 / 0.28, 62.9}};
```

```
In[*]:= datafit3 = NonlinearModelFit[datafrecond3, {v * a + b}, {v, b}, a]
      |ajusta a modelo no lineal
```

```
Out[*]:=
```

```
FittedModel[[-0.04 + 17.5392 a]]
```

```
In[*]:= datafit3[{"BestFit", "ParameterTable"}]
```

```
Out[*]:=
```

	Estimate	Standard Error	t-Statistic	P-Value
$-0.04 + 17.5392 a$				
v	17.5392	0.169434	103.516	5.22207×10^{-8}
b	-0.04	0.392768	-0.101841	0.923784

```
In[*]:= (*m=2kg*)
```

```
In[*]:= datafrecond4 = {{1 / 1.68, 11.8}, {1 / 0.84, 24},
      {1 / 0.56, 34.6}, {1 / 0.42, 48.9}, {1 / 0.336, 60.4}, {1 / 0.28, 72.5}};
```

```
In[*]:= datafit4 = NonlinearModelFit[datafrecond4, {v * a + b}, {v, b}, a]
      |ajusta a modelo no lineal
```

```
Out[*]:=
```

```
FittedModel[[-0.666667 + 20.496 a]]
```

```
In[*]:= datafit4[{"BestFit", "ParameterTable"}]
```

```
Out[*]:=
```

	Estimate	Standard Error	t-Statistic	P-Value
$-0.666667 + 20.496 a$				
v	20.496	0.318336	64.3849	3.48592×10^{-7}
b	-0.666667	0.73794	-0.903416	0.417393

```
In[*]:= (*Gráficas*)
```

```
graph1 = Plot[datafit1[a], {a, 0, 5}, PlotLegends → {"Cuerda 1, 2 kg"},
      |representación gráfica |leyendas de representación
      BaseStyle → {FontFamily → "Calibri", FontSize → 14},
      |estilo base |familia de tipo de letra |tamaño de tipo de letra
      PlotStyle → {Blend[{Blue, White}], Thick}, PlotRange → {{0.2, 5}, {5, 105}},
      |estilo de repre... |mezcla... |azul |blanco |grueso |rango de representación
      Frame → True, FrameLabel → {Style["λ-1 / m-1", Bold, 16], Style["ν / Hz", 16, Bold]}};
      |marco |verd... |etiqueta de marco |estilo |negrita |estilo |negrita
```

```
In[*]:= pplot1 = ListPlot[datafrecond1, PlotStyle → {Black, Thickness[0.003]}};
      |representación de lista |estilo de repre... |negro |grosor
```

```
In[*]:= Needs["ErrorBarPlots`"];
      |necesita
```

```
(*dataerr1=ErrorListPlot[
      {{1/1.68,12.4},ErrorBar[0.00708,0.1]},{{1/0.84,25.2},ErrorBar[0.014,0.1]},
      {{1/0.56,36.5},ErrorBar[0.0212,0.1]},{{1/0.42,46.3},ErrorBar[0.0283,0.1]},
      {{1/0.336,57.1},ErrorBar[0.035,0.1]},{{1/0.28,73.2},ErrorBar[0.0425,0.1]},
      {{1/0.24,86.1},ErrorBar[0.0496,0.1]},{{1/0.21,101.9},ErrorBar[0.0566,0.1]}},
      AxesOrigin→{0,0},PlotStyle→{Black,Thickness[0.003]}};*)
      |origen de ejes |estilo de repre... |negro |grosor
```

```
In[*]:= dataerr1 = ErrorListPlot[
      {{1 / 0.28, 73.2}, ErrorBar[0.0425, 0.1]}, {1 / 0.24, 86.1}, ErrorBar[0.0496, 0.1]},
      {{1 / 0.21, 101.9}, ErrorBar[0.0566, 0.1]}},
      AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.003]}};
      |origen de ejes |estilo de repre... |negro |grosor
```

```

In[*]:= graph5 = Plot[datafit5[a], {a, 0, 5}, PlotLegends → {"Cuerda 1, 1.5 kg"},
  representación gráfica      leyendas de representación
  BaseStyle → {FontFamily → "Calibri", FontSize → 14},
  estilo base      familia de tipo de letra      tamaño de tipo de letra
  PlotStyle → {Blend[{Red, White}], Thick}, PlotRange → {{0, 5}, {0, 105}},
  estilo de repre... mezcla... rojo blanco grueso rango de representación
  Frame → True, FrameLabel → {Style["λ-1 / m-1", Bold, 16], Style["ν / Hz", 16, Bold]};
  marco      verd... etiqueta de marco estilo      negrita      estilo      negrita

```

```

In[*]:= pplot5 = ListPlot[datafrecond5,
  representación de lista
  PlotLegends → {"Datos experimentales"}, PlotStyle → {Black, Thickness[0.003]};
  leyendas de representación      estilo de repre... negro grosor

```

```

In[*]:= (*Los errores en la cuerda 1, 1.5 kg, no se aprecian*)

```

```

In[*]:= (*dataerr5=ErrorListPlot[{ {1/1.68,11.3},ErrorBar[0.00708,0.1]},
  { {1/0.84,20.8},ErrorBar[0.014,0.1]}, { {1/0.56,28.7},ErrorBar[0.0212,0.1]}},
  AxesOrigin→{0,0},PlotStyle→{Black,Thickness[0.003]};*)
  origen de ejes      estilo de repre... negro grosor

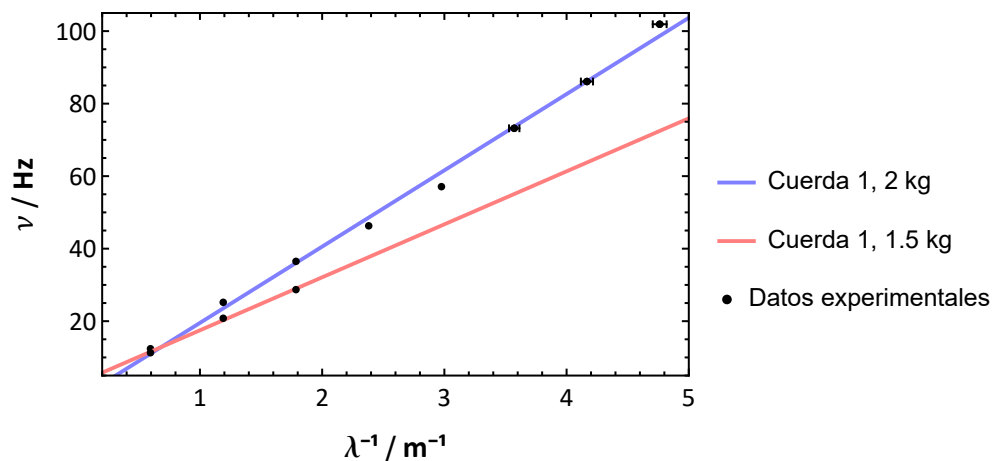
```

```

In[*]:= Show[graph1, pplot1, dataerr1, graph5, pplot5]
  muestra

```

Out[*]=



```

In[*]:= graph2 = Plot[datafit2[a], {a, 0, 5}, PlotLegends → {"Cuerda 2, 1 kg"},
  representación gráfica      leyendas de representación
  BaseStyle → {FontFamily → "Calibri", FontSize → 14},
  estilo base      familia de tipo de letra      tamaño de tipo de letra
  PlotStyle → {Blend[{Red, White}], Thick}, PlotRange → {{0.2, 4}, {5, 80}},
  estilo de repre... mezcla... rojo blanco grueso rango de representación
  Frame → True, FrameLabel → {Style["λ-1 / m-1", Bold, 16], Style["ν / Hz", 16, Bold]};
  marco      verd... etiqueta de marco estilo      negrita      estilo      negrita

```

```

In[*]:= pplot2 = ListPlot[datafrecond2, PlotStyle → {Black, Thickness[0.003]};
  representación de lista      estilo de repre... negro grosor

```

```

In[*]:= dataerr2 = ErrorListPlot[
  {{ {1 / 0.42, 35.5}, ErrorBar[0.0283, 0.1]}, { {1 / 0.336, 42.5}, ErrorBar[0.0354, 0.1]}},
  { {1 / 0.28, 54.1}, ErrorBar[0.0425, 0.1]}},
  AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.003]};
  origen de ejes      estilo de repre... negro grosor

```

```

In[*]:= graph3 = Plot[datafit3[a], {a, 0, 5}, PlotLegends → {"Cuerda 2, 1.5 kg"},
  [representación gráfica] [leyendas de representación]
  BaseStyle → {FontFamily → "Calibri", FontSize → 14},
  [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
  PlotStyle → {Blend[{Blue, White}], Thick}, PlotRange → {{0.2, 5}, {5, 105}},
  [estilo de repre...] [mezcla...] [azul] [blanco] [grueso] [rango de representación]
  Frame → True, FrameLabel → {Style[" $\lambda^{-1}$  / m $^{-1}$ ", Bold, 16], Style[" $\nu$  / Hz", 16, Bold]}};
  [marco] [verd...] [etiqueta de marco] [estilo] [negrita] [estilo] [negrita]

In[*]:= pplot3 = ListPlot[datafrecond3, PlotStyle → {Black, Thickness[0.003]};
  [representación de lista] [estilo de repre...] [negro] [grosor]

In[*]:= dataerr3 = ErrorListPlot[
  {{1 / 0.42, 42.1}, ErrorBar[0.0283, 0.1]}, {1 / 0.336, 51.7}, ErrorBar[0.0354, 0.1]},
  {1 / 0.28, 62.9}, ErrorBar[0.0425, 0.1]}},
  AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.003]};
  [origen de ejes] [estilo de repre...] [negro] [grosor]

In[*]:= graph4 = Plot[datafit4[a], {a, 0, 5}, PlotLegends → {"Cuerda 2, 2 kg"},
  [representación gráfica] [leyendas de representación]
  BaseStyle → {FontFamily → "Calibri", FontSize → 14},
  [estilo base] [familia de tipo de letra] [tamaño de tipo de letra]
  PlotStyle → {Blend[{Black, White}], Thick}, PlotRange → {{0.2, 5}, {5, 105}},
  [estilo de repre...] [mezcla...] [negro] [blanco] [grueso] [rango de representación]
  Frame → True, FrameLabel → {Style[" $\lambda^{-1}$  / m $^{-1}$ ", Bold, 16], Style[" $\nu$  / Hz", 16, Bold]}};
  [marco] [verd...] [etiqueta de marco] [estilo] [negrita] [estilo] [negrita]

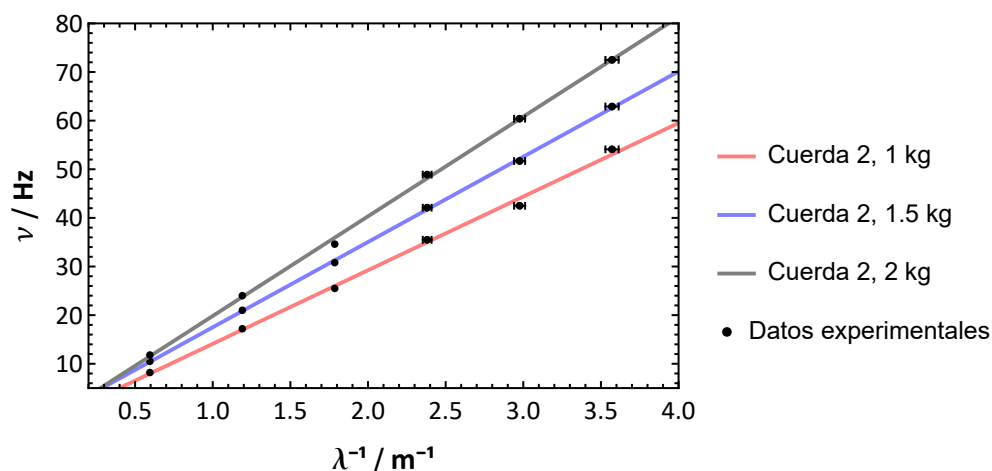
In[*]:= pplot4 = ListPlot[datafrecond4,
  [representación de lista]
  PlotLegends → {"Datos experimentales"}, PlotStyle → {Black, Thickness[0.003]};
  [leyendas de representación] [estilo de repre...] [negro] [grosor]

In[*]:= dataerr4 = ErrorListPlot[
  {{1 / 0.42, 48.9}, ErrorBar[0.0283, 0.1]}, {1 / 0.336, 60.4}, ErrorBar[0.0354, 0.1]},
  {1 / 0.28, 72.5}, ErrorBar[0.0425, 0.1]}},
  AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.003]};
  [origen de ejes] [estilo de repre...] [negro] [grosor]

In[*]:= Show[graph2, pplot2, dataerr2, graph3, pplot3, dataerr3, graph4, pplot4, dataerr4]
  [muestra]

```

Out[*]=



```

In[*]:= (*Cálculo de las sigmas y rhos, densidades*)

```

```

In[*]:=  $\sigma[\tau\_ , v\_ ] := \tau / v^2$ ;

In[*]:=  $\epsilon\sigma[\tau\_ , v\_ , \epsilon v\_ ] := (2 * \tau / v^3) * \epsilon v$ ;

In[*]:=  $\sigma[\{14.7, 19.6, 9.8, 14.7, 19.6\}, \{14.616, 21.038, 15.1392, 17.5392, 20.496\}]$ 

Out[*]=
{0.0688114, 0.044284, 0.0427583, 0.0477857, 0.0466571}

In[*]:=  $\epsilon\sigma[\{14.7, 19.6, 9.8, 14.7, 19.6\},$ 
 $\{14.616, 21.038, 15.1392, 17.5392, 20.496\}, \{0.7759, 0.64136, 0.403, 0.1694, 0.318\}]$ 

Out[*]=
{0.00730579, 0.00270007, 0.00227642, 0.000923063, 0.00144779}

In[*]:=  $\rho[\sigma\_ , R\_ ] := \sigma / (\text{Pi} * R^2)$ ;
[número pi]

In[*]:=  $\epsilon\rho[R\_ , \sigma\_ , \epsilon R\_ , \epsilon\sigma\_ ] := \text{Sqrt}[(\epsilon\sigma / (\text{Pi} * R^2))^2 + (((2 * \sigma) / (\text{Pi} * R^3)) * \epsilon R)^2]$ ;
[raíz cuadrada [número pi [número pi]

In[*]:=  $\rho[\{0.06881137975073766, 0.044284033416153216, 0.04275827961134217,$ 
 $0.047785680382456695, 0.046657111290274424\}, \{0.00115, 0.00115, 0.001, 0.001, 0.001\}]$ 

Out[*]=
{16562.1, 10658.6, 13610.4, 15210.7, 14851.4}

In[*]:=  $\epsilon\rho[\{0.00115, 0.00115, 0.001, 0.001, 0.001\},$ 
 $\{0.06881137975073766, 0.044284033416153216, 0.04275827961134217, 0.047785680382456695,$ 
 $0.046657111290274424\}, \{0.000025, 0.000025, 0.000025, 0.000025, 0.000025\},$ 
 $\{0.007305794957388801, 0.0027000672755760078, 0.002276419716150245,$ 
 $0.0009230631108360886, 0.001447790924112731\}]$ 

Out[*]=
{1900.15, 798.181, 994.063, 815.316, 873.951}

In[*]:= (*Cálculo de frec 1 teórica*)

In[*]:=  $f1[v\_ , \lambda1\_ ] := v / \lambda1$ ;

In[*]:=  $f1[\{14.616, 21.038, 15.1392, 17.5392, 20.496\}, \{1.68, 1.68, 1.68, 1.68, 1.68\}]$ 

Out[*]=
{8.7, 12.5226, 9.01143, 10.44, 12.2}

In[*]:=  $\epsilon f1[v\_ , \lambda1\_ , \epsilon v\_ , \epsilon\lambda1\_ ] := \text{Sqrt}[(\epsilon v / \lambda1)^2 + ((v / \lambda1^2) * \epsilon\lambda1)^2]$ 
[raíz cuadrada]

In[*]:=  $\epsilon f1[\{14.616, 21.038, 15.1392, 17.5392, 20.496\}, \{1.68, 1.68, 1.68, 1.68, 1.68\},$ 
 $\{0.776, 0.641, 0.4, 0.17, 0.318\}, \{0.02, 0.02, 0.02, 0.02, 0.02\}]$ 

Out[*]=
{0.473374, 0.409638, 0.261148, 0.16027, 0.238586}

In[*]:= (*densidades medias*)

In[*]:=  $\sigma_{med1} = (68.811379 + 44.28403) / 2$ 

Out[*]=
56.5477

```

```
In[*]:= eomed1 = Sqrt[ (1 / 2) * ( (68.811379 - 56.5477045) ^ 2 + (44.28403 - 56.5477045) ^ 2) ]
      raíz cuadrada
```

```
Out[*]=
12.2637
```

```
In[*]:= omed2 = (42.75827961134217 + 47.785680382456695 + 46.657111290274424) / 3
```

```
Out[*]=
45.7337
```

```
In[*]:= eomed2 =
      Sqrt[ (1 / (3 * 2)) * ( (45.73369042802443 - 42.75827961134217) ^ 2 + (45.73369042802443 -
      raíz cuadrada
      47.785680382456695) ^ 2 + (45.73369042802443 - 46.657111290274424) ^ 2) ]
```

```
Out[*]=
1.52296
```

```
In[*]:= pomed1 = (16.562073691196147 + 10.658635641931882) / 2
```

```
Out[*]=
13.6104
```

```
In[*]:= epomed1 = Sqrt[ (1 / 2) * ( (13.610354666564014 - 16.562073691196147) ^ 2 +
      raíz cuadrada
      (13.610354666564014 - 10.658635641931882) ^ 2) ]
```

```
Out[*]=
2.95172
```

```
In[*]:= pomed2 = (13.610383116501023 + 15.21065448375479 + 14.851419784471707) / 3
```

```
Out[*]=
14.5575
```

```
In[*]:= epomed2 =
      Sqrt[ (1 / 6) * ( (14.557485794909173 - 13.610383116501023) ^ 2 + (14.557485794909173 -
      raíz cuadrada
      15.21065448375479) ^ 2 + (14.557485794909173 - 14.851419784471707) ^ 2) ]
```

```
Out[*]=
0.484773
```